ARJUN BHATTACHARYA, PHD

Assistant Professor of Epidemiology · MD Anderson Cancer Center

The Bhattacharya Lab for Computational Genomics

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APPOINTMENTS

Assistant Professor, Department of Epidemiology 2023–
Affiliate, Institute for Data Science in Oncology
UT MD Anderson Cancer Center · Houston, TX

Fellow 2021–23
UCLA Institute for Quantitative & Computational Biosciences

Postdoctoral Fellow 2020–23
Pathology & Laboratory Medicine, UCLA · Los Angeles, CA

EDUCATION

PhD, Biostatistics 2015–20
University of North Carolina at Chapel Hill

BS, Mathematical Decision Sciences · BS, Biology 2011–15
University of North Carolina at Chapel Hill
Mackenzie Family Foundation Innovation Scholar

GRANT FUNDING

NIH/NIEHS R21 ES039048 PI — **$491,915**
Identifying PFAS-induced disruption of placental imprinting and neurodevelopment through multi-omic negative control calibration
09/2026 – 08/2028

NIH/NCI R21 CA293419 PI — **$402,000**
Alternative splicing and isoform expression as mediators for the genetic etiology of breast cancer
07/2024 – 06/2026

UT System Rising STARs Award PI — **$109,000**
Start-up support for computational genomics research
06/2024

MD Anderson CCSG New Faculty Award PI — **$50,000**
Cancer Center Support Grant pilot funding
04/2025

AWARDS & HONORS

- **2025** Emerging Leader, *British Journal of Cancer*
- **2025–** Academic Editor, *PLOS Genetics*
- **2024** UT Rising STARs Award
- **2023** UCLA QCB Collaboratory Service Award
- **2022** Semifinalist, Epstein Postdoctoral Award, ASHG (top 10%)
- **2022** James V. Neel Award, International Genetic Epidemiology Society
- **2022** Most Outstanding Oral Presentation, World Congress of Psychiatric Genetics
- **2021** O'Malley Alumni Award for Publication Excellence in Population Sciences
- **2021** UCLA Bruins in Genomics Outstanding Mentorship Award
- **2021–23** ISPG Early Career Investigator Program Award
- **2021–23** Fellow, UCLA Institute for Quantitative and Computational Biosciences
- **2021** SER Conference Scholarship
- **2019–20** Center for Environmental Health and Susceptibility Training Grant
- **2018–19** Susan G. Komen Graduate Training Fellowship in Breast Cancer Disparities
- **2017–18** UNC-CH Department of Biostatistics Tuition Award
- **2011–15** Mackenzie Family Foundation Innovation Scholarship (full, 4 years)

PUBLICATIONS

42 PEER-REVIEWED

A Bhattacharya · trainee · * equal contribution · ★ first/senior author or trainee-led

2026

1 ★ **Quantification method affects replicability of eQTL analysis, colocalization, and TWAS**
ST Head, ST Bresnahan, Y Chang, V Shah, N Cole, WH Wu, **A Bhattacharya**
Nature Genetics · 2026 ↗ doi.org/10.1038/s41588-026-02827-0

2 ★ **Natural variation in transplacental transfer efficiency exposes distinct transcriptional network architectures of PFAS effects on birth weight and gestational age**
ST Bresnahan, HEJ Yong, MG Drelichman, SN Campbell, AE Trapse, GR Romo, CM Cellini, S Lopez, JK Chan, S Chan, ER Elkin, **A Bhattacharya***, JY Huang*
Environmental Health Perspectives · 2026 ↗ doi.org/10.1021/ehp.6c00646

3 **Comparative 10-year performance of mammography artificial intelligence, polygenic, and clinical breast cancer risk models in the Kaiser Permanente Research Bank**
V Arasu, …, ST Head, **A Bhattacharya**, …, W Sieh
Journal of the National Cancer Institute · 2026 ↗ doi.org/10.1093/jnci/djag154

4 ★ **Long-read transcriptome assembly reveals vast isoform diversity in the placenta associated with metabolic and endocrine function**
ST Bresnahan, H Yong, WH Wu, S Lopez, J Kok Yen Chan, F White, P Jacques, M Hivert, S Chang, MI Love, JY Huang*, **A Bhattacharya***
Nature Communications · 2026 ↗ doi.org/10.1038/s41467-026-71303-4

2025

5 ★ **Isoform-level analyses of 6 cancers uncover extensive genetic risk mechanisms undetected at the gene-level**
Y Chang, ST Head, T Harrison, Y Yu, C Huff, B Pasaniuc, S Lindström, **A Bhattacharya**
British Journal of Cancer · 2025 Emerging Leaders in Cancer Special Edition ↗ doi.org/10.1038/s41416-025-03141-y

2024

6 **Splicing-specific transcriptome-wide association uncovers genetic mechanisms for schizophrenia**
J Hervoso, K Amoah, J Dodson, M Choudhury, **A Bhattacharya**, G Quinones-Valdez, B Pasaniuc, G Xiao
American Journal of Human Genetics · 2024 PMID: 38925119 ↗ doi.org/10.1016/j.ajhg.2024.06.001

7 **Enhanced CTLA-4 blockade anti-tumor immunity with APG-157 combination in a murine head and neck cancer**
D Shin, S Basak, M Veena, B Comin-Anduix, **A Bhattacharya**, T Dong, A Ko, P Han, J Jacobs, N Moatamed, L Avila, M Pellegrini, M Wang, E Srivatsan
Cancer Medicine · 2024 PMID: 38686626 ↗ doi.org/10.1002/cam4.7212

8 **Cross-ancestry, cell-type-informed atlas of gene, isoform, and splicing regulation in the developing human brain**
C Wen, M Margolis, R Dai, P Zhang, P Przytycki, D Vo, **A Bhattacharya**, N Matoba, C Jiao, M Kim, E Tsai, C Hoh, N Aygün, R Walker, C Chatzinakos, D Clarke, H Pratt, PsychENCODE Consortium, M Peters, M Gerstein, N Daskalakis, Z Weng, A Jaffe, J Kleinman, T Hyde, D Weinberger, N Bray, N Sestan, D Geschwind, K Roeder, A Gusev, B Pasaniuc, J Stein, M Love, K Pollard, C Liu*, M Gandal*
Science · 2024 PMID: 38781368 ↗ doi.org/10.1126/science.adh0829

9 **Brain cell-type shifts in Alzheimer's disease, autism, and schizophrenia interrogated using methylomics and genetics**
C Yap, D Voi, M Heffel, **A Bhattacharya**, C Wen, K Keeper, J Zheng, Z Zheng, E Hannon, D Vellame, A Franklin, C Caggiano, J Mill, N Zaitlen, A Gusev, B Pasaniuc, C Lou, M Gandal
Science Advances · 2024 PMID: 38781333 ↗ doi.org/10.1126/sciadv.adn7655

10 **Genome-wide association meta-analysis identifies novel ancestry-specific primary open-angle glaucoma loci and shared biology with vascular mechanisms and cell proliferation**
V Lo Faro, **A Bhattacharya**, W Zhou, D Zhou, Y Wang, K Läll, M Kanai, E Lopera-Maya, P Straub, P Pawar, R Tao, X Zhong, S Namba, the Global Biobank Meta-analysis Initiative, S Sanna, IM Nolte, Y Okada, N Ingold, S MacGregor, H Sneider, I Surakka, C Willer, AR Martin, MA Brantley Jr, ER Gamazon, NM Jansonius, K Joos, NJ Cox, J Hirbo
Cell Reports Medicine · 2024 PMID: 38382466 ↗ doi.org/10.1016/j.xcrm.2024.101430

11 ★ **Polygenic scores for tobacco use provide insights into systemic health risks in a diverse EHR-linked biobank in Los Angeles**
V Venkateswaran, K Boulier, Y Ding, R Johnson, **A Bhattacharya**[*], B Pasaniuc[*]
Translational Psychiatry · 2024 PMID: 38238290 ↗ doi.org/10.1038/s41398-024-02743-z

2023 12 ★ **Isoform-level transcriptome-wide association uncovers extensive novel genetic risk mechanisms for neuropsychiatric disorders in the human brain**
A Bhattacharya, D Vo, C Jops, M Kim, C Wen, J Hervoso, B Pasaniuc[*], M Gandal[*]
Nature Genetics · 2023 PMID: 38036788 ↗ doi.org/10.1038/s41588-023-01560-2

13 **Multi-ancestry genome-wide association study of cannabis use disorder yields insight into disease biology and public health implications**
D Levey, M Galimberti, J Deak, F Wendt, **A Bhattacharya**, D Koller, K Harrington, R Quaden, E Johnson, M Cooke, V Rajagopal, S Empke, H Zhou, Y Nunez, H Kranzler, H Edenberg, A Agrawal, J Smoller, D Demontis, VA Million Veteran Program, J Gaziano, M Gandal, R Polimanti, M Stein, J Gelernter
Nature Genetics · 2023 PMID: 37985822 ↗ doi.org/10.1038/s41588-023-01563-z

14 **Genotype error due to low-coverage sequencing induces uncertainty in polygenic scoring**
E Petter, Y Ding, K Hou, **A Bhattacharya**, A Gusev, N Zaitlen, B Pasaniuc
American Journal of Human Genetics · 2023 PMID: 37490908 ↗ doi.org/10.1016/j.ajhg.2023.06.015

15 **Impact of cross-ancestry genetic architecture on GWAS in admixed populations**
R Mester, K Hou, Y Ding, G Meeks, K Burch, **A Bhattacharya**, B Henn, B Pasaniuc
American Journal of Human Genetics · 2023 PMID: 37224807 ↗ doi.org/10.1016/j.ajhg.2023.05.001

16 **twas_sim, a Python-based tool for simulation and power analysis of transcriptome-wide association analysis**
X Wang, Z Lu, **A Bhattacharya**, B Pasaniuc, N Mancuso
Bioinformatics · 2023 PMID: 37099718 ↗ doi.org/10.1093/bioinformatics/btad288

17 **Causal effects on complex traits are similar across segments of different continental ancestries within admixed individuals**
K Hou, Y Ding, Z Xu, A Wu, **A Bhattacharya**, R Mester, G Belbin, J Cai, D Conti, M Fornage, C Gignoux, X Guo, C Haiman, R Kaplan, E Kenny, M Kim, C Kooperberg, L Lange, A Manichaikul, K North, N Nudelman, U Peters, L Rasmussen-Torvik, S Rich, J Rotter, H Wheeler, Y Zhou, the PAGE Consortium, S Sankararaman, B Pasaniuc
Nature Genetics · 2023 PMID: 36941441 ↗ doi.org/10.1038/s41588-023-01338-6

18 **The UCLA ATLAS Community Health Initiative: promoting precision health research in a diverse biobank**
R Johnson, Y Ding, **A Bhattacharya**, A Chiu, C Lajonchere, D Geschwind, B Pasaniuc
Cell Genomics · 2023 PMID: 36777178 ↗ doi.org/10.1016/j.xgen.2022.100243

2022 19 **Broad transcriptomic dysregulation occurs across the cerebral cortex in ASD**
M Gandal[*], J Haney[*], B Wamsley, C Yap, S Parhami, P Emani, N Chang, G Chen, G Hoftman, D de Alba, G Ramaswami, C Hartl, **A Bhattacharya**, C Luo, T Jin, D Wang, R Kawaguchi, D Quintero, J Ou, Y Wu, N Parikshak, V Swarup, T Belgard, M Gerstein, B Pasaniuc, D Geschwind
Nature · 2022 PMID: 36323788 ↗ doi.org/10.1038/s41586-022-05377-7

20 ★ **Best practices of multi-ancestry, meta-analytic transcriptome-wide association studies: lessons from the Global Biobank Meta-analysis Initiative**
A Bhattacharya[*], J Hirbo[*], D Zhou, W Zhou, J Zheng, M Kanai, the Global Biobank Meta-analysis Initiative, B Pasaniuc, E Gamazon, N Cox
Cell Genomics · 2022 PMID: 36341024 ↗ doi.org/10.1016/j.xgen.2022.100180

21 **Proteome-wide Mendelian randomization in global biobank meta-analysis reveals trans-ancestry drug targets for common diseases**
H Zhao, H Rasheed, T Nost, Y Cho, Y Liu, L Bhatta, **A Bhattacharya**, the Global Biobank Meta-analysis Initiative, G Hemani, G Davey Smith, B Brumpton[*], W Zhou[*], B Neale[*], T Gaunt[*], J Zheng[*]
Cell Genomics · 2022 PMID: 36388766 ↗ doi.org/10.1016/j.xgen.2022.100195

22 **Meta-analysis fine-mapping is often miscalibrated at single-variant resolution**
M Kanai, R Elzur, W Zhou, the Global Biobank Meta-analysis Initiative
Cell Genomics · 2022 Consortium author ↗ doi.org/10.1016/j.xgen.2022.100210

23 **Global Biobank analyses provide lessons for developing polygenic risk scores across diverse cohorts**
Y Wang, S Namba, E Lopera, S Kerminen, K Tsuo, K Läll, M Kanai, W Zhou, K Wu, M Favé, L Bhatta, P Awadalla, B Brumpton, P Deelen, K Hveem, V Lo Faro, R Mägi, Y Murakami, S Sanna, J Smoller, J Uzunovic, B Wolford, the Global Biobank Meta-analysis Initiative, C Willer, E Gamazon, N Cox, I Surakka, Y Okada, A Martin, J Hirbo
Cell Genomics · 2022 Consortium author ↗ doi.org/10.1016/j.xgen.2022.100241

24 **Leveraging global multi-ancestry meta-analysis in the study of idiopathic pulmonary fibrosis genetics**
J Partanen, P Häppölä, W Zhou, A Lehisto, M Ainola, E Sutinen, R Allen, A Stockwell, O Leavy, J Oldham, B Guillen-Guio, N Cox, J Hirbo, D Schwartz, T Fingerlin, C Flores, I Noth, B Yaspan, R Jenkins, L Wain, S Ripatti, M Pirinen, International IPF Genetics Consortium, the Global Biobank Meta-analysis Initiative, T Laitinen, R Kaarteenaho, M Myllärniemi, M Daly, J Koskela
Cell Genomics · 2022 Consortium author ↗ doi.org/10.1016/j.xgen.2022.100181

25 **A practical guideline of genomics-driven drug discovery in the era of global biobank meta-analysis**
S Namba, T Konuma, K Wu, W Zhou, the Global Biobank Meta-analysis Initiative, Y Okada
Cell Genomics · 2022 Consortium author · PMID: 36643910 ↗ doi.org/10.1016/j.xgen.2022.100190

26 **Multi-ancestry meta-analysis of asthma identifies novel associations and highlights the value of increased power and diversity**
K Tsuo, W Zhou, Y Wang, M Kanai, S Namba, R Gupta, L Majara, L Nkambule, T Morisaki, Y Okada, B Neale, the Global Biobank Meta-analysis Initiative, M Daly, A Martin
Cell Genomics · 2022 Consortium author · PMID: 36778051 ↗ doi.org/10.1016/j.xgen.2022.100212

27 **Global Biobank Meta-analysis Initiative: powering genetic discovery across human diseases**
W Zhou, M Kanai, K Wu, R Humaira, K Tsuo, J Hirbo, Y Wang, **A Bhattacharya**, …, C Willer, M Daly, B Neale for the Global Biobank Meta-analysis Initiative
Cell Genomics · 2022 PMID: 36777996 ↗ doi.org/10.1016/j.xgen.2022.100192

28 **Leveraging genomic diversity for discovery in an EHR-linked biobank: the UCLA ATLAS Community Health Initiative**
R Johnson, Y Ding, V Venkateswaran, **A Bhattacharya**, A Chiu, T Schwarz, M Freund, L Zhan, K Burch, C Caggiano, B Hill, N Rakocz, B Balliu, J Sul, N Zaitlen, V Arboleda, E Halperin, S Sankararaman, M Butte, UCLA Precision Health Data Discovery Repository Working Group, UCLA Precision Health ATLAS Working Group, C Lajonchere, D Geschwind, B Pasaniuc
Genome Medicine · 2022 PMID: 36085083 ↗ doi.org/10.1186/s13073-022-01106-x

29 **Innovative computational approaches shed light on genetic mechanisms underlying cognitive impairment among children born extremely preterm**
W Liu, L Huang, Q Sun, **A Bhattacharya**, X Tan, K Kuban, R Joseph, T O'Shea, R Fry, Y Li, H Santos
Journal of Neurodevelopmental Disorders · 2022 IDDRC Special Issue · PMID: 35240980 ↗ doi.org/10.1186/s11689-022-09429-x

30 ★ **Placental genomics mediates genetic associations with complex health traits and disease**
A Bhattacharya, A Freedman, V Avula, R Harris, W Liu, Y Li, R Joseph, L Smeester, H Hartwell, K Kuban, T O'Shea, C Marsit, R Fry, H Santos
Nature Communications · 2022 PMID: 35121757 ↗ doi.org/10.1038/s41467-022-28365-x

31 ★ **Gene-level germline contributions to clinical risk of recurrence scores in Black and White breast cancer patients**
A Patel, M García-Closas, A Olshan, C Perou, M Troester, M Love, **A Bhattacharya**
Cancer Research · 2022 PMID: 34711612 ↗ doi.org/10.1158/0008-5472.CAN-21-1207

32 **Racial differences in breast cancer outcomes by hepatocyte growth factor pathway expression**
G Jones, KA Hoadley, H Benefield, L Olsson, A Hamilton, **A Bhattacharya**, E Kirk, H Tipaldos, J Fleming, K Williams, M Love, H Nichols, A Olshan, M Troester
Breast Cancer Research and Treatment · 2022 PMID: 35034243 ↗ doi.org/10.1007/s10549-021-06497-w

2021 33 **On powerful GWAS in admixed populations**
K Hou, **A Bhattacharya**, R Mester, K Burch, B Pasaniuc
Nature Genetics · 2021 PMID: 34824480 ↗ doi.org/10.1038/s41588-021-00953-5

34 **Hepatocyte growth factor pathway expression in breast cancer by race and subtype**
G Jones, K Hoadley, L Olsson, A Hamilton, **A Bhattacharya**, E Kirk, H Tipaldos, J Fleming, M Love, H Nichols, A Olshan, M Troester
Breast Cancer Research · 2021 PMID: 34344422 ↗ doi.org/10.1186/s13058-021-01460-5

35 ★ **DeCompress: tissue compartment deconvolution of targeted mRNA expression panels using compressed sensing**
A Bhattacharya, A Hamilton, M Troester, M Love
Nucleic Acids Research · 2021 PMID: 33524140 ↗ doi.org/10.1093/nar/gkab031

36 ★ **MOSTWAS: Multi-Omic Strategies for Transcriptome-Wide Association Studies**
A Bhattacharya, Y Li, M Love
PLOS Genetics · 2021 PMID: 33684137 ↗ doi.org/10.1371/journal.pgen.1009398

2020 37 **Biopsychosocial correlates of psychological distress in Latina mothers**
H Santos, H Adynski, R Harris, **A Bhattacharya**, A Incollingo-Rodriguez, R Cali, A Torres Yabar, B Nephew, C Murgatroyd
Journal of Affective Disorders · 2020 PMID: 33445084 ↗ doi.org/10.1016/j.jad.2020.12.193

38 ★ **Evidence for the placenta-brain axis: multi-omic kernel aggregation predicts intellectual and social impairment in children born extremely preterm**
H Santos*, **A Bhattacharya***, R Joseph, L Smeester, K Kuban, C Marsit, T O'Shea, R Fry
Molecular Autism · 2020 PMID: 33308293 ↗ doi.org/10.1186/s13229-020-00402-w

39 ★ **An approach for normalization and quality control for NanoString RNA expression data**
A Bhattacharya*, A Hamilton*, M Troester, K Hoadley, M Love
Briefings in Bioinformatics · 2020 PMID: 32789507 ↗ doi.org/10.1093/bib/bbaa163

40 ★ **A framework for transcriptome-wide association studies in breast cancer in diverse study populations**
A Bhattacharya, M García-Closas, A Olshan, C Perou, M Troester, M Love
Genome Biology · 2020 PMID: 32079541 · 2021 O'Malley Alumni Award for Publication Excellence ↗ doi.org/10.1186/s13059-020-1942-6

2019 41 **Epigenome-wide DNA methylation in placentas from preterm infants: association with maternal socioeconomic status**
H Santos, **A Bhattacharya**, E Martin, K Addo, M Psioda, L Smeester, R Joseph, S Hooper, J Frazier, K Kuban, T O'Shea, R Fry
Epigenetics · 2019 PMID: 31062658 ↗ doi.org/10.1080/15592294.2019.1614743

2018 42 **Discrimination exposure and DNA methylation of stress-related genes in Latina mothers**
H Santos, B Nephew, **A Bhattacharya**, X Tan, L Smith, R Alyamani, E Martin, K Perreira, R Fry, C Murgatroyd
Psychoneuroendocrinology · 2018 PMID: 30144780 ↗ doi.org/10.1016/j.psyneuen.2018.08.014

PREPRINTS & SUBMITTED 8

2026 1 ★ **ICONIC: An R package for integrating instrumental variable- and negative-control-informed causal discovery and diagnostics in multiomic studies**
ST Bresnahan, C Xiong, ST Head, Y Chang, **A Bhattacharya***, JY Huang*
medRxiv · 2026 ↗ doi.org/10.64898/2026.08.26.26361466

2 ★ **Improving isoform-level eQTL and integrative genetic analyses of breast cancer risk with long-read RNA transcript assemblies**
ST Head, A Nemani, Y Chang, TA Harrison, ST Bresnahan, JH Rothstein, W Sieh, S Lindström, **A Bhattacharya**
bioRxiv · 2026 In review, Nature Communications ↗ doi.org/10.64898/2026.03.22.713514

3 **Cell-type aware transcriptome-wide association study of mammographic density phenotypes**
A Sistig*, JH Rothstein*, S Zhu, ST Head, Y Chang, N Achacoso, SE Alexeeff, V Arasu, T Gadgil, LD Gerstley, LR Margolies, LC Sakoda, L Shen, CLS Gueye, M Villaseñor, M Westley, MODE/BCAC Consortia, **A Bhattacharya**, RJ Klein, LA Habel, X Song, P Wang, W Sieh
medRxiv · 2026 In review, Nature Communications ↗ doi.org/10.1101/2025.11.25.25341027

4 **Integrative proteogenomic analyses reveal insights into subtype-specific glioma risk**
T Nakase, K Smith-Byrne, Q Ostrom, K Suhre, B Melin, M Rentoft, C Ogamba, G Henderson, C Carlo, **A Bhattacharya**, ST Head, G Guerra, M Wrensch, J Wiencke, R Jenkins, J Eckel-Passow, M Bondy, S Francis, L Kachuri
Submitted · 2026 In review, Nature Communications

5 ★ **Formal statistical replication in lung cancer genome-wide association studies**
Y Chang, J Byun, BR Gorman, RJ Huang, JD McKay, CI Amos, S Pyarajan, **A Bhattacharya***, R Sun*
medRxiv · 2026 In review, HGG Advances ↗ doi.org/10.1101/2025.10.02.25337130

6 ★ **Distal gene regulation mediated by non-coding RNAs contributes to germline risk for breast and prostate cancer**
N Cole, P Lee, T Schwarz, P Zhang, M Freedman, A Gusev, S Lindström, M Gandal, B Pasaniuc, **A Bhattacharya**
medRxiv ↗ doi.org/10.1101/2022.02.08.22270601

2023 7 ★ **Interplay of serum bilirubin and tobacco smoking with lung and head and neck cancers in a diverse, EHR-linked Los Angeles biobank**
V Venkateswaran, E Petter, K Boulier, Y Ding, **A Bhattacharya***, B Pasaniuc*
medRxiv · 2023 PMID: 37873378 ↗ doi.org/10.1101/2023.09.29.23296364

2022 8 **Multivariate variance components analysis uncovers genetic architecture of brain isoform expression and novel psychiatric disease mechanisms**
M Kim, D Vo, C Jops, C Wen, A Patowary, **A Bhattacharya**, C Yap, H Zhou, M Gandal
medRxiv · 2022 ↗ doi.org/10.1101/2022.10.18.22281204

SELECTED & INVITED TALKS

39

Improved measurement and methods to understand placental mechanisms mediating maternal exposures on fetal outcomes
Baylor OB/GYN Research Roundtable — Jan 2027
INVITED

Opportunities and advancements in integrating long- and short-read RNA-seq across differential expression analysis, QTL mapping, and outcome prediction
Joint Statistical Meetings, Boston — Aug 2026
INVITED

Splicing-resolved genomics for precision women's health across populations
Lunenfeld-Tanenbaum Research Institute, University of Toronto — Jan 2026
INVITED

Discovering novel isoform regulation in complex diseases by integrating long- and short-read RNA-sequencing
Institute for Molecular Medicine Finland (FIMM), HiLIFE, University of Helsinki — Oct 2025
INVITED

Isoform-specific analyses to uncover hidden cancer risk mechanisms
Novel Statistical Approaches for Studying Multi-omics Data, Banff International Research Station — Jul 2025
INVITED

Pacific Center for Genome Research, University of Hawaii Cancer Center — Mar 2025
INVITED

UNC Biostatistics 75th Anniversary Celebration — Nov 2024
INVITED

Vanderbilt University Division of Epidemiology — Sep 2024
INVITED

Integrating environment, gene expression, & computational models to identify modifiable pathways across diverse populations
Pediatric Academic Societies Annual Meeting — Apr 2025
STATE-OF-THE-ART PLENARY

Considerations for computational molecular precision medicine
MD Anderson Department of General Internal Medicine — Jul 2024
INVITED

MD Anderson Division of Cancer Prevention and Control Grand Rounds — Jun 2024
INVITED

Integrative computational methods to unlock the wealth of information in bulk RNA-sequencing
American Statistical Association STATGEN 2024 — May 2024

UNC-Chapel Hill BIOS Genomics Seminar — Jan 2024
INVITED

UTHealth Houston SPH Human Genetics Center — Jan 2024
INVITED

Isoform-level TWAS uncover novel biological mechanisms underlying genetic associations with neuropsychiatric traits
American College of Neuropsychopharmacology Annual Meeting — Dec 2023
INVITED PANEL

UPenn Symposium, Genetics and Epigenetics of Behavior — May 2023
INVITED

ASHG Annual Meeting — Oct 2022
PLATFORM · EPSTEIN AWARD SEMIFINALIST

Harvard Program in Genetic Epidemiology and Statistical Genetics — Oct 2022
INVITED

World Congress of Psychiatric Genetics — Sep 2022
ORAL · MOST OUTSTANDING ORAL PRESENTATION

International Genetic Epidemiology Society — Sep 2022
ORAL · JAMES V. NEEL AWARD

Gene-level germline associations with breast cancer subtype and survival

Breast Ovarian Consortia Meeting — Jun 2023
INVITED

Leveraging the breadth and depth of transcriptomics to understand the genetic etiology of complex traits

NYU Center of Human Genetics and Genomics — Feb 2023
INVITED

MD Anderson Department of Cancer Epidemiology — Jan 2023
INVITED

Emory School of Medicine Human Genetics — Jan 2023
INVITED

UNC Computational Medicine — Jan 2023
INVITED

USC Keck Center of Genetic Epidemiology — Dec 2022
INVITED

Carnegie Mellon Computational Biology — Dec 2022
INVITED

Lieber Institute for Brain Development — Nov 2022
INVITED

Regeneron Genetics Center, Statistical Genetics — Oct 2022
INVITED

Distal mediator-enriched, placental TWAS of 40 traits suggest genetic mechanisms supporting the DOHaD hypothesis

Vanderbilt University Medical Center Division of Epidemiology — Dec 2022
INVITED

US DOHaD Society Annual Meeting — Nov 2021
JUNIOR INVESTIGATOR KEYNOTE

ASHG Annual Meeting — Oct 2021
PLATFORM

World Congress of Psychiatric Genetics — Oct 2021
ORAL · ECIP

Singapore Institute for Clinical Sciences — Aug 2021
INVITED

Society for Epidemiologic Research — Jun 2021
ORAL

UCLA QCBio Research Seminar — Apr 2021
INVITED

DeCompress: tissue compartment deconvolution for targeted RNA panels using compressed sensing

ICCABS, Computational Advances for NGS Workshop — Dec 2020
INVITED

MOSTWAS: Multi-Omic Strategies for Transcriptome-Wide Association Studies

Society for Epidemiologic Research — Dec 2020
ORAL

ASHG Annual Meeting — Oct 2020
PLATFORM

MENTORSHIP

POSTDOCTORAL SCHOLARS

Taylor Head, PhD
Current

Sean Bresnahan, PhD
Current

GRADUATE STUDENTS

Yung-Han Chang
PhD, Quantitative Sciences, MDA GSBS · co-advised with Ryan Sun

Yan Zhang
PhD, Quantitative Sciences, MDA GSBS · co-advised with Callie Kwartler · AHA Predoctoral Fellow

Jacob Smith
CPRTP Summer Program

Achal Patel
PhD, Epidemiology, UNC → Data Scientist, Genentech

Vidhya Venkateswaran
PhD, Oral Biology, UCLA → Senior Research Fellow & NIH DATA Scholar, NIDCR

UNDERGRADUATES

Aryun Nemani
Rice University

Charis Xiong
Rice University

Vihan Shah
UCSD

William Wu
Rice → MD, UT Southwestern (full scholarship)

Erika Vasquez
Rice → Research Associate, NIDCR

Nolan Cole
UCLA BIG → Biostatistics PhD, U. Washington

Paige Lee
UCLA BIG → MS Data Science, Harvard

SERVICE & TEACHING

EDITORIAL & REVIEW

Academic Editor, *PLOS Genetics*

Grant review: NIH NIDA ZDA1 DRO-N (Member); Swiss National Science Foundation

Referee: Nature, Nature Genetics, Nature Biotechnology, Nature Communications, AJHG, eLife, PLOS Genetics

Abstract review: SER Annual Meetings; ASHG 2023

Reviewer, MD Anderson NCI Summer Research Trainee Program, 2024

ACADEMIC SERVICE

MDA GSBS Student Success & Well-being Committee — Member 2024–, **Chair 2026–**

TEACHING

UCLA QCB W5, RNA-seq Analysis · 2021–2023

BIOS 735, Introduction to Data Science · 2019

BIOS 550, Basic Elements of Probability & Statistical Inference · 2018

BIOS 673, Probability & Statistics · 2017

COMPUTING

Advanced: R (incl. package development), SAS, LaTeX, HPC

Intermediate: Python, C++, MATLAB

OPEN-SOURCE SOFTWARE

`isotwas`
Isoform-level transcriptome-wide association studies

`MOSTWAS`
Distal-mediator enriched TWAS

`DeCompress`
Cell-type deconvolution for targeted RNA panels

`NanoNormIter`
QC & normalization for NanoString nCounter